Tahsin Tajwar Tanni

Dhaka, Bangladesh | tahsintanni009@gmail.com | (+880) 1796518460

Portfolio | **LinkedIn** | **Github** | **Google Scholar**

Profile

I'm a Computer Science and Engineering graduate from BRAC University with experience spanning AI automation, machine learning research, and full-stack web development. My interests lie in artificial intelligence, machine learning, agentic systems, full-stack development, blockchain, and building impactful technologies that solve real-world problems.

Skills

- **Programming Languages:** Python, JavaScript, TypeScript, C, Solidity
- **Databases:** MySQL, MongoDB, SQLite, Prisma ORM
- **Frameworks & Libraries:** React.js, Next.js, Node.js, Express.js, NumPy, pandas, scikit-learn, PyTorch, Matplotlib, Seaborn.
- **AI & Automation:** OpenAI API, Anthropic API, n8n, RAG, Prompt Engineering
- **CSS Frameworks:** Tailwind CSS
- **Tools & Platforms:** Git, GitHub, Postman, Hugging Face Spaces, Docker.

Education

BRAC University, BSc in Computer Science and Engineering — 2022 – February, 2026
- GPA: 3.60 / 4.00

Govt. Haji Muhammad Mohsin College, Higher Secondary Certificate (HSC) — 2018 – 2020
- GPA: 5.00 / 5.00

Bangladesh Mohila Samity Girls' High School and College, Secondary School Certificate (SSC) — 2004 – 2018
- GPA: 5.00 / 5.00

Experience

AI Automation Trainee, 180 RE, Toronto, Canada — May 2026 – Present
- Built LLM-powered workflows using OpenAI and Anthropic APIs, automating multi-step business processes via n8n and third-party service integrations.
- Developed and deployed AI-powered chatbots, integrating them into websites to automate user interactions and support.
- Applied prompt engineering techniques to optimise LLM outputs for accuracy and safety across use cases.

Python Tutor, Zentorra — February 2025 – May 2025
- Tutored high school students in Python programming, covering fundamentals through advanced topics, and assisted with coding assignments and projects.

Junior Web Developer (Part-time), Weabers — February 2023 – July 2024
- Developed and maintained responsive web applications, building reusable UI components and integrating APIs with backend teams.
- Debugged issues, refactored code, and deployed feature updates across production codebases.

Junior Executive, IT Department, Robotics Club of BRAC University — June 2022 – August 2023
- Collaborated with team members to manage internal digital platforms and technical resources.

Projects

RAG-Slackbot
- A Slack-integrated Retrieval-Augmented Generation (RAG) chatbot built with n8n that enables real-time querying of internal PDF knowledge bases.
- Performs automated document ingestion, text chunking, vector storage (Supabase), and context-aware response generation using Gemini, with confidence-based escalation and source attribution.
- Tools Used: n8n, Supabase, Ollama, Google Gemini, Slack API

ProfMail – AI-Powered Academic Cold Email Generator
- An AI-driven web application that automates personalized cold email generation for graduate school applicants, featuring real-time professor lab scraping and research overlap analysis between student thesis and faculty publications.
- Includes customizable tone and length controls with an interactive editing workspace and local email history.
- Tools Used: React, TypeScript, Vite 6, Tailwind CSS

AI Professor Finder
- An automated n8n workflow that discovers and evaluates potential MSc/PhD supervisors using Google Scholar and Gemini AI.
- Reads professor information from Google Sheets, retrieves recent publications via SerpAPI, compares research against a student's thesis, and produces AI-generated compatibility scores and funding recommendations.
- Tools Used: n8n, Google Scholar, SerpAPI, Gemini AI, Google Sheets

Realtime Collaborative Playlist

- Built a real-time collaborative music playlist application with Server-Sent Events (SSE) for live synchronization, drag-and-drop reordering using fractional indexing, a voting system, and now-playing simulation.
- Tools Used: Next.js 16, React 19, TypeScript, Prisma ORM, SQLite, Tailwind CSS 4, Framer Motion

Thesis-Management-System-ThesisFlow

- This project Thesis Management System is a digital platform that streamlines thesis submission, tracking, and supervision. It enhances communication, organization, and efficiency in managing academic research projects
- Tools Used: MongoDB, Express.js, React, Node.js, HTML, CSS

Stock Forecasting Analysis Toolkit

- Developed an interactive Hugging Face Space demonstrating data synthesis using generative AI techniques for job-related analytics.
- Enabled dynamic user interaction for synthetic data generation and visualization through a web-based interface.
- Tools Used: pandas, numpy, matplotlib, statsmodels, scikit-learn, prophet, tensorflow, gradio, Hugging Face

vae-dec-clustering-uncertainty

- This project is the implementation and analysis of a Variational Autoencoder (VAE) combined with Deep Embedded Clustering (DEC) for unsupervised image clustering and also explores uncertainty quantification (via entropy and latent variance) to evaluate clustering correctness and model confidence.
- Tools Used: python, pytorch, pandas, NumPy, scikit-learn, matplotlib, seaborn

Research Experience

Undergraduate Researcher January 2025 – January 2026
BRAC University, Department of Computer Science and Engineering

- Investigated post-deployment vulnerabilities in pretrained language models – weight-noise injection and low-precision quantization attacks that degrade stability without altering architecture.
- Designed **Anchor-Guided Repair**, a fine-tuning defense jointly optimizing language modeling loss and anchor regularization to constrain parameter drift from a clean baseline.
- Validated across multiple LLM architectures and attack types, consistently recovering performance close to the clean baseline.

Publication: Rohan, Khan, **Tanni**, Fardin, & Bushra (2026). Anchor-Guided Repair: A Defense Mechanism for Enhancing Stability of Compromised Pretrained Language Models. *IndabaX Nigeria 2026, PMLR 319*, 368–381.

Phantom Guard: Fine-tuned Multi-Agent System for Offline Slopsquatting Detection Ongoing
Role: Researcher

- Designing a dual-LLM generator-reviewer pipeline with registry/API fallback validation to detect hallucinated, slopsquatted package recommendations.
- Exploring knowledge distillation for lightweight, low-latency self-verifying models.

Achievement

- Achieved **2nd Runner-up** at the **CSE471 Project Demonstration** with *ThesisFlow*, a comprehensive **Thesis Management System** designed to centralize and streamline the entire thesis lifecycle at BRAC University, featuring role-based dashboards, progress tracking, structured feedback, and AI-powered academic guidance.
- Successfully completed an **Intermediate Web Development** course during the Residential Semester at BRAC University, gaining practical skills in designing responsive websites.
- Completed the **Intro to Machine Learning** course on Kaggle, gaining foundational knowledge in data science and machine learning with practical applications to real-world problems.
- Completed **AI Prompt Engineer™ (AI CERTs™)** via AgentX Bangladesh (Netcom Learning & Microsoft). Skills: advanced prompt design (zero / few shots, chain of thought), PFSET / PAIPS,IPS, XML steering, agent workflows, and ethical AI use.